

Integer Factorisation Problem

Input $\rightarrow$ Integer: $N \geq 2$

Output $\rightarrow$ Prime Factorisation of N

Greatest Common Divisor / G.C.D. Problem

Input $\rightarrow N, M : N, M \in \text{Whole Numbers}$

Output $\rightarrow \text{gcd}(N, M)$

Measuring Computational Cost

Input $\rightarrow$ Computation $\rightarrow$ Output

$\rightarrow$ Encoding and Input Length

Number	Encoding	Length
0	0	1
1	1	1
2	10	2
3	11	2
4	100	3
5	101	3

$$lg(N) = \begin{cases} 1 & \text{if } N=0 \\ 1 + \log_2 N & \text{if } N \geq 1 \end{cases}$$

$\downarrow$
length of binary
encoding of N

Integer Factorisation

Simple trial division $\rightarrow \text{Size}(C_n) = O(n^2 \cdot 2^{n/2})$

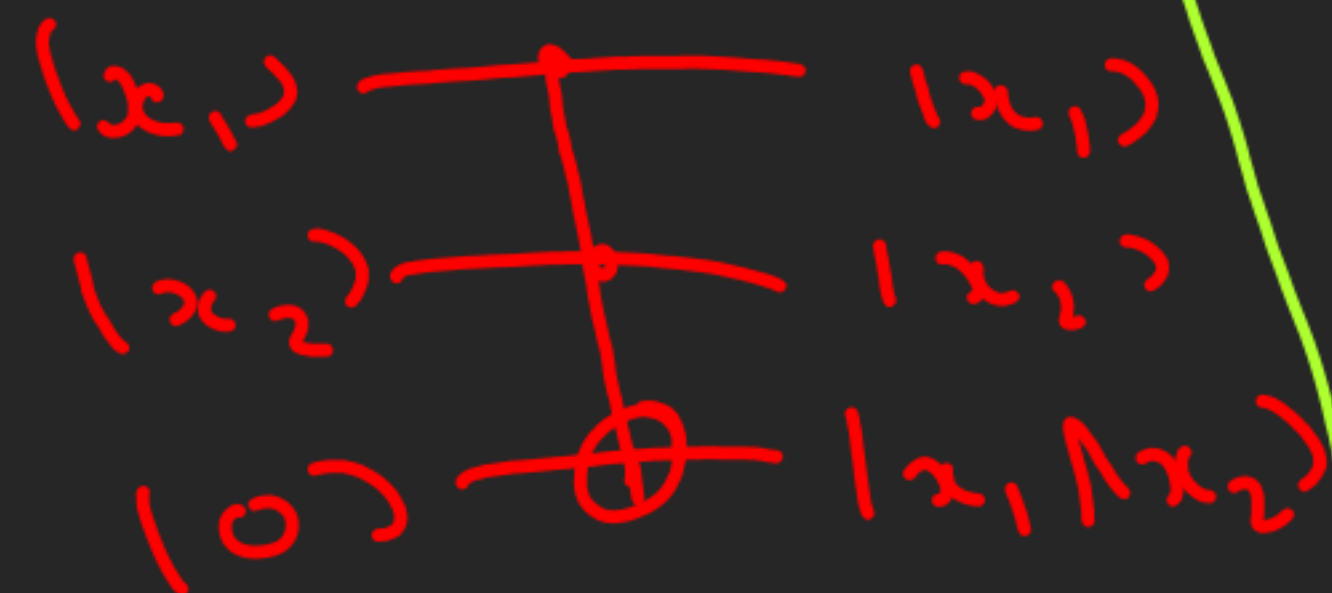
Number Field Sieve $\rightarrow \text{Size}(C_n) = 2^{O(n^{1/3} \cdot (\log^{2/3}(n)))}$

Classical to Quantum Circuits

$\rightarrow$ We can build a quantum circuit at approximately the same cost as that of a classical circuit



Quantum AND



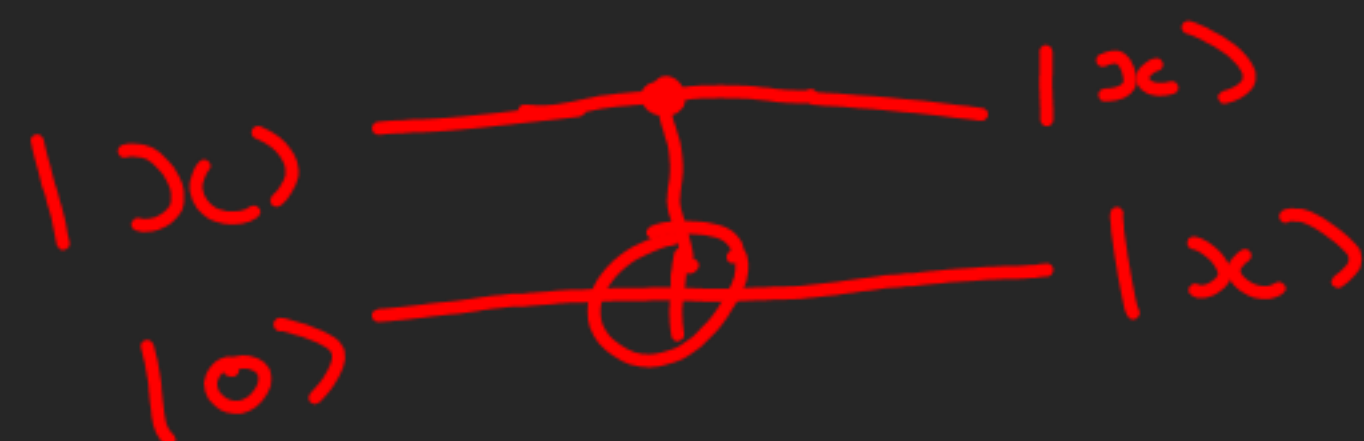
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